

Salesforce Project

Scholarship Application Management System

1. Project Overview & Abstract

The **Scholarship Application Management System** is a cloud-based application developed on the **Salesforce Lightning Platform**. It automates the scholarship distribution lifecycle across higher education institutions. The system tracks student credentials, configures merit/need-based scholarship criteria, enforces strict eligibility rules using declarative logic, and delivers real-time analytical visibility via executive reports and dashboards.

2. Data Model & Architecture

The system utilizes a relational data model with three custom objects:

- **Student (Student__c)**: Parent master entity capturing candidate academic and economic details.
- **Scholarship Scheme (Scholarship_Scheme__c)**: Parent entity defining grant awards, income ceilings, and minimum CGPA criteria.
- **Scholarship Application (Scholarship_Application__c)**: Junction entity linking students to applicable schemes via a Master-Detail relationship with Student and a Lookup relationship with Scholarship Scheme.

3. Object & Field Implementation

F9.. Quick Find New Deleted Fields Field Dependencies Set History Tracking				
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIE...	INDEXED
Annual Family Income	Annual_Family_Income__c	Currency(10, 2)		☐
CGPA	CGPA__c	Number(2, 2)		☐
Created By	CreatedById	Lookup(User)		
Department	Department__c	Picklist		☐
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		☒
Roll Number	Roll_Number__c	Text(20)		☐
Student Email	Student_Email__c	Email		☐
Student Name	Name	Text(80)		☒

Student Object Configuration

The **Student** object stores persistent candidate data. Key custom fields configured include:

- **Roll Number (Roll_Number__c)**: Text(20), marked as Unique to eliminate duplicate submissions.
- **Student Email (Student_Email__c)**: Email field for notifications and communications.
- **Department (Department__c)**: Picklist defining academic disciplines (CSE, IT, ECE, EEE, MECH, CIVIL).
- **CGPA (CGPA__c)**: Number(2, 2) recording cumulative grade points for eligibility audits.
- **Annual Family Income (Annual_Family_Income__c)**: Currency field validating economic need thresholds.

Scholarship Scheme Object Configuration

F9

Quick Find

New

Deleted Fields

Field Dependencies

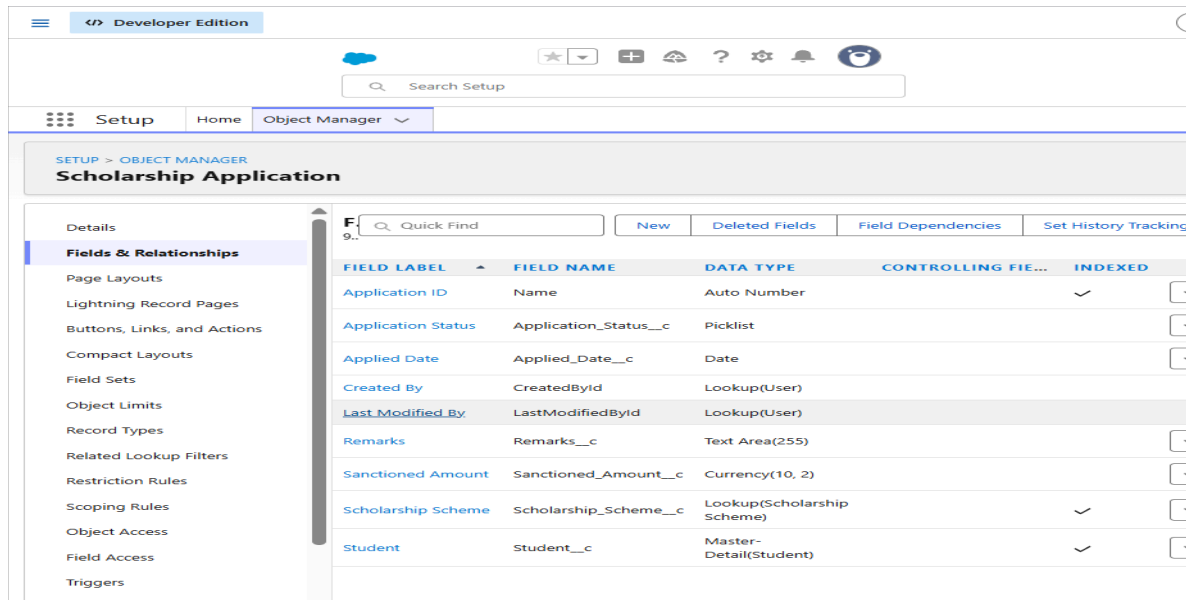
Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIE...	INDEXED
Category	Category__c	Picklist		
Created By	CreatedById	Lookup(User)		
Deadline	Deadline__c	Date		
Last Modified By	LastModifiedById	Lookup(User)		
Max Eligible Income	Max_Eligible_Income__c	Currency(10, 2)		
Min CGPA Required	Min_CGPA_Required__c	Number(2, 2)		
Owner	OwnerId	Lookup(User,Group)	✓	
Scholarship Amount	Scholarship_Amount__c	Currency(10, 2)		
Scholarship Scheme Name	Name	Text(80)	✓	

The **Scholarship Scheme** object serves as the catalog for all institutional grants. Key fields configured include:

- **Scholarship Amount (Scholarship_Amount__c)**: Currency field denoting the allocated financial award.
- **Min CGPA Required (Min_CGPA_Required__c)**: Minimum academic threshold for selection.
- **Max Eligible Income (Max_Eligible_Income__c)**: Upper ceiling for need-based schemes.
- **Deadline (Deadline__c)**: Date field preventing applications beyond expiry.
- **Category (Category__c)**: Picklist classifying schemes into Merit-Based, Need-Based, First Graduate, and Sports.

Scholarship Application (Junction Object)



This junction object associates candidates with target schemes while enforcing cascade deletes if a student record is removed.

- **Student (Student__c):** Master-Detail Relationship to Student__c.
- **Scholarship Scheme (Scholarship_Scheme__c):** Lookup Relationship to Scholarship_Scheme__c.
- **Application ID (Name):** Auto-Number field formatted as APP- {0000}.
- **Application Status (Application_Status__c):** Picklist representing workflow stages (Draft, Submitted, Under Review, Approved, Rejected).
- **Sanctioned Amount (Sanctioned_Amount__c):** Actual grant sanctioned post-verification.

4. Custom Application & User Experience

New Lightning App

App Details

* App Name ⓘ

Scholarship Management Syst

* Developer Name ⓘ

Scholarship_Management_Syst

Description ⓘ

System to manage student scholarship applications and schemes.

App Branding

Image ⓘ



APPLY FOR SCHOLARSHIPS

Clear

Primary Color Hex Value ⓘ

#76CA69

Org Theme Options

☐ Use the app's image and color instead of the org's custom theme

App Launcher Preview



Scholarship Management S...
System to manage student scholarship applications and ...

A custom Lightning App titled **Scholarship Management System** was provisioned via the App Manager. The navigation environment brings all business entities together under unified navigation tabs (**Students**, **Scholarship Schemes**, **Scholarship Applications**, **Reports**, and **Dashboards**). Profile-level security was configured to grant full access privileges to the System Administrator.

New Lightning App

Available Items

Create

Type to filter list...

Customers

Customizable Pr...

Data Action Targ...

Data Actions

Data Detect

Data Explorer

Data Graphs

Data Lake Objects

Data Manager H...

Data Model

Selected Items

Students

Scholarship Schemes

Scholarship Applicat...

Reports

Dashboards

5. Declarative Automation: Validation Rule

To maintain data integrity and enforce academic criteria automatically, a validation rule was configured on the junction object.

Rule Name: Check_CGPA_Eligibility

Object: Scholarship_Application__c

Student__r.CGPA__c < Scholarship_Scheme__r.Min_CGPA_Required__c

- **Error Message:** Eligibility Error: Student's CGPA is less than the minimum CGPA required for this scholarship scheme!

SETUP > OBJECT MANAGER

Scholarship Application

Scholarship Application Validation Rule

Define a validation rule by specifying an error condition and a corresponding error message. The error condition is written as a Boolean formula expression that returns true or false. When the formula expression returns true, the save will be aborted and the error message will be displayed. The user can correct the error and try again.

Validation Rule Edit [Save] [Save & New] [Cancel]

Rule Name: Check_CGPA_Eligibility

Active: ☒

Description: Prevents application if student CGPA is less than minimum CGPA required.

Error Condition Formula ⓘ = Required Information

Example: Discount_Percent__c > 0.30 [More Examples...](#)
Display an error if Discount is more than 30%

If this formula expression is true, display the text defined in the Error Message area

[Insert Field] [Use Formula Assistant] [Insert Operator ▼]

Student__r.CGPA__c < Scholarship_Scheme__r.Min_CGPA_Required__c

[Check Syntax] No errors found

Quick Tips

- [Operators & Functions](#)

Functions

-- All Function Categories

- ABS
- ACOS
- ADDMONTHS
- AND
- ASCII
- ASIN

[Insert Selected Function]

ABS(number)
Returns the absolute value of a number, a number without a sign.

[Help on this function](#)

Einstein

6. Functional Testing & Error Handling

NEW STUDENT

* = Required Information

Information

* Student Name

Madhumitha DS

Roll Number

23ISR028

Student Email

madzshanmugam@gmail.com

Department

MSC

CGPA

9.13

Annual Family Income

180000

Owner

MADHUMITHA DS

Cancel

Save & New

Save

NEW STUDENT

* = Required Information

Information

* Student Name

Karthik R

Roll Number

21IT15

Student Email

karthik@example.com

Department

IT

CGPA

7.40

Annual Family Income

150000

Owner

MADHUMITHA DS

Cancel

Save & New

Save

\$180,000.00

Information

Application ID

* Student

Madhumitha DS

Scholarship Scheme

National Merit Scholars

Application Status

Submitted

Applied Date

9/10/2026

Sanctioned Amount

\$50,000.00

Remarks

All documents verified

Cancel

Save & New

Save

* = Required In

Information

Application ID

* Student

Karthik R

Scholarship Scheme

First Graduate Financial

Application Status

Under Review

Applied Date

9/11/2026

Sanctioned Amount

\$35,000.00

Remarks

Documents verified, eligible for aid.

Cancel

Save & New

Explanation: Functional test cases executed:

- **Positive Flow:** A record (APP-0001) with a student CGPA of 8.90 applying for a scheme with a minimum requirement of 8.50 committed successfully.
- **Negative Flow:** An application with an ineligible CGPA triggered the runtime error banner, preventing invalid record creation and confirming rule integrity.

7. Operational Analytics (Reports)

Category

Recently Used

All

Accounts & Contacts

Opportunities

Customer Support Reports

Leads

Campaigns

Activities

Contracts and Orders

Price Books, Products and Assets

Administrative Reports

File and Content Reports

Create Report

Select a Report Type

Showing results for Scholarship Applications

Report Type Name

Students with Scholarship Applications

Students with Scholarship Applications and Scholarship Scheme

Details

Students with Scholarship Appli...
Standard Report Type

Start Report

Details

Fields (43)

Owner

Scholarship Scheme

Student

Role

Developer Edition

Scholarship Management ...

Students

Scholarship Schemes

Scholarship Applications

Reports

Dashboards

REPORT

New Report

Students with Scholarship Applications and Scholarship Scheme

Previewing a limited number of records. Run the report to see everything.

Student: Student Name

Scholarship Application: Application ID

Scholarship Scheme: Scholarship Scheme Name

1	Karthik R	APP-0002	First Graduate Financial Aid
2	Madhumitha DS	APP-0001	National Merit Scholarship

Outline

Filters

Groups

GROUP ROWS

Add group...

Columns

Add column...

Student: Student Name

Scholarship Application: Application ID

Scholarship Scheme: Scholarship Scheme Name

Developer Edition

Scholarship Management ...

Students

Scholarship Schemes

Scholarship Applications

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REPORT

New Report

Students with Scholarship Applications and Scholarship Scheme

Previewing a limited number of records. Run the report to see everything.

Application Status

Submitted

Under Review

Record Count

Submitted

Under Review

Submitted (1)	Madhumitha DS	APP-0001	National Merit Scholarship
Under Review (1)	Karthik R	APP-0002	First Graduate Financial Aid
Total (2)			

Outline

Filters

Groups

GROUP ROWS

Add group...

GROUP COLUMNS

Add group...

Columns

Add column...

Student: Student Name

Scholarship Application: Application ID

Scholarship Scheme: Scholarship Scheme Name

X

Save Report

Applications by Status

Report Unique Name ⓘ

Applications_by_Status_THo

Report Description

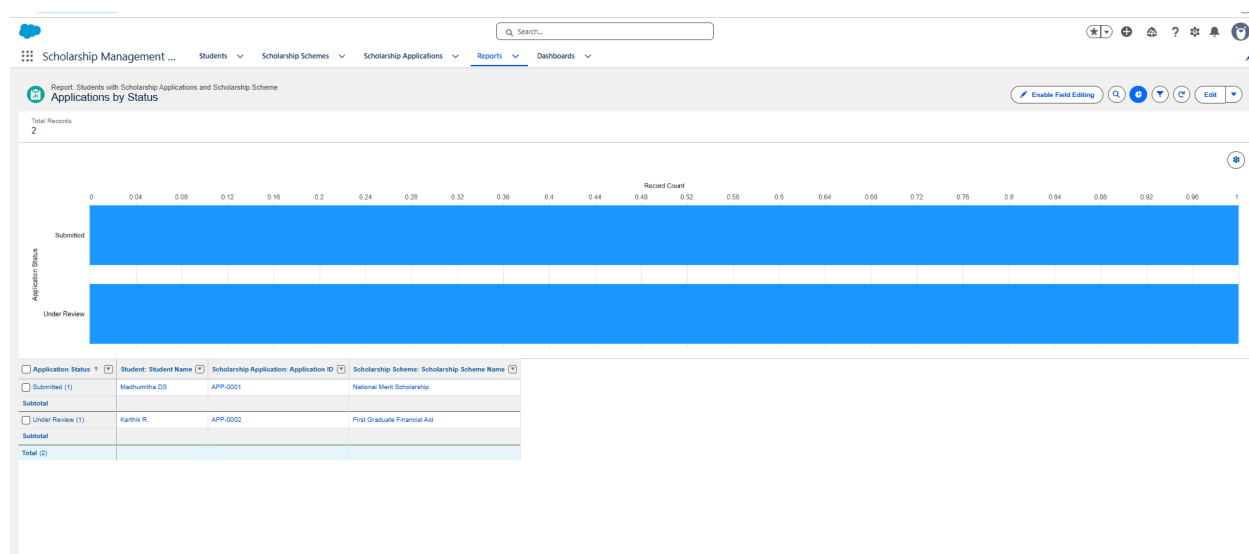
Folder

Private Reports

Select Folder

Cancel

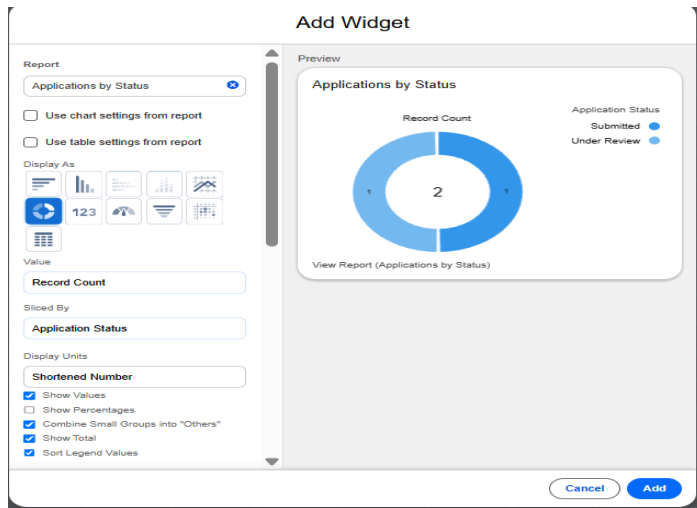
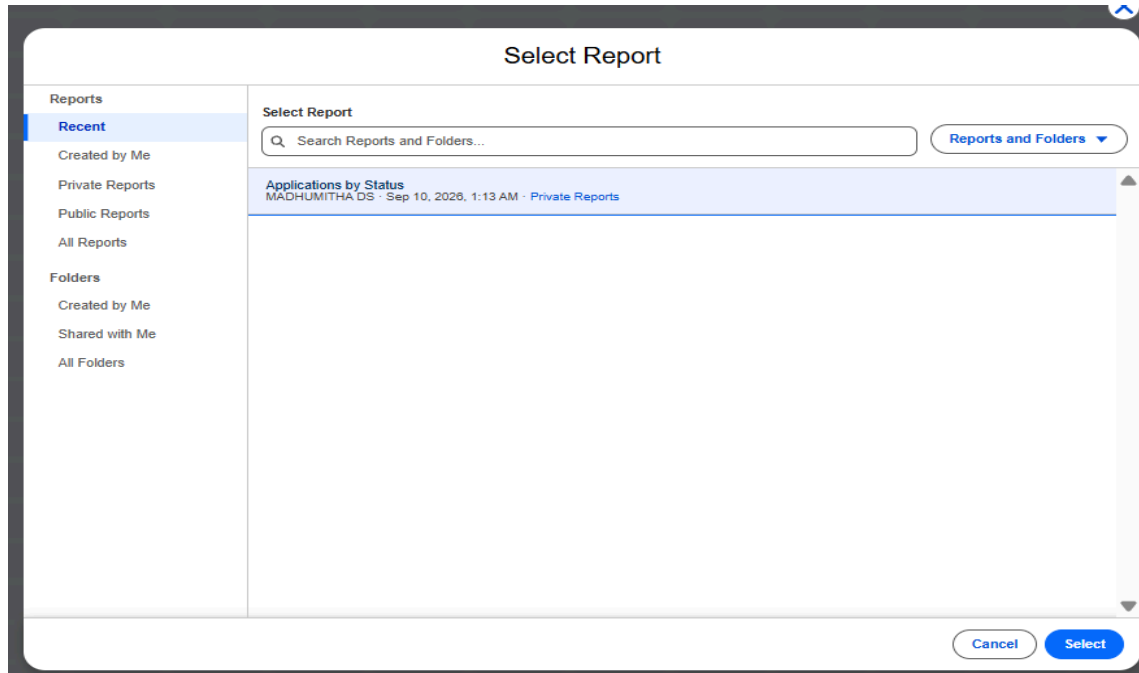
Save

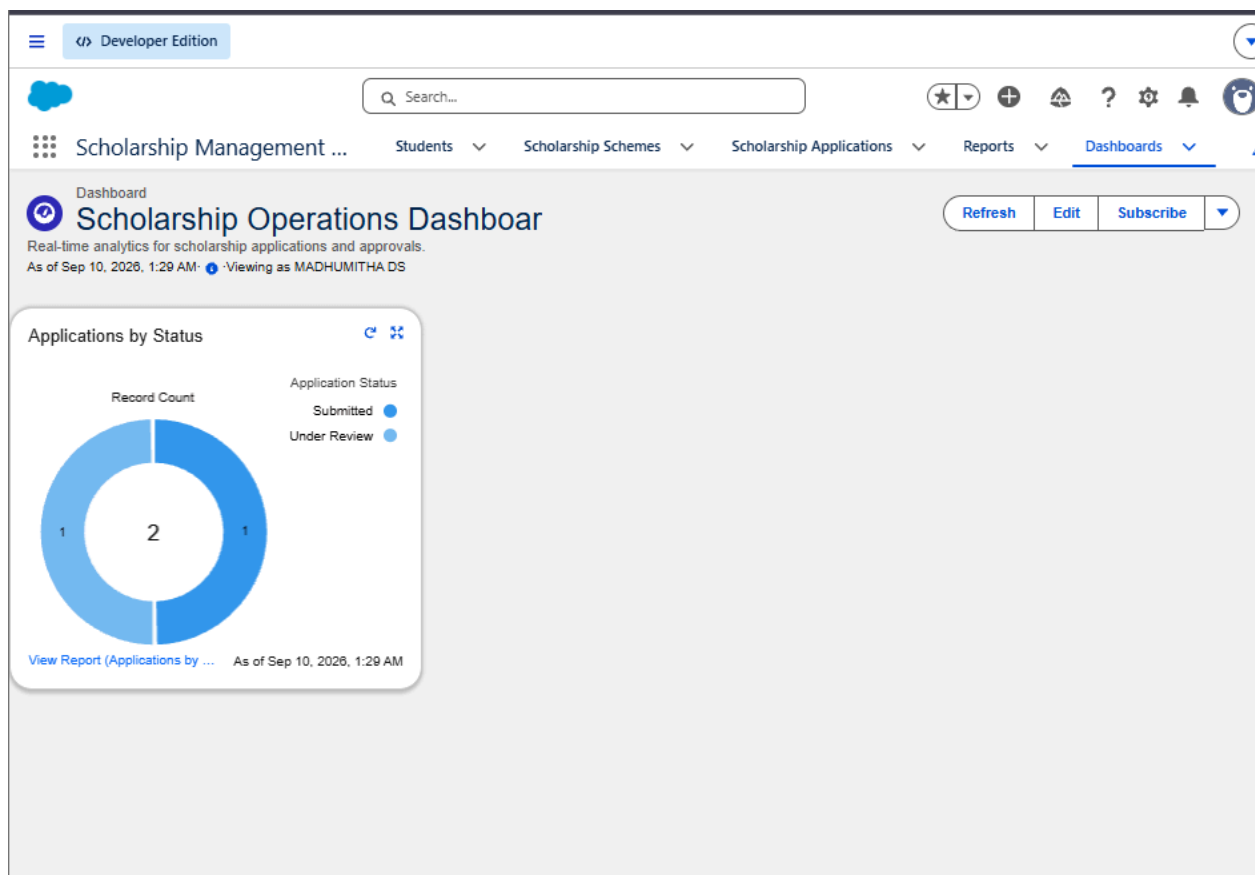
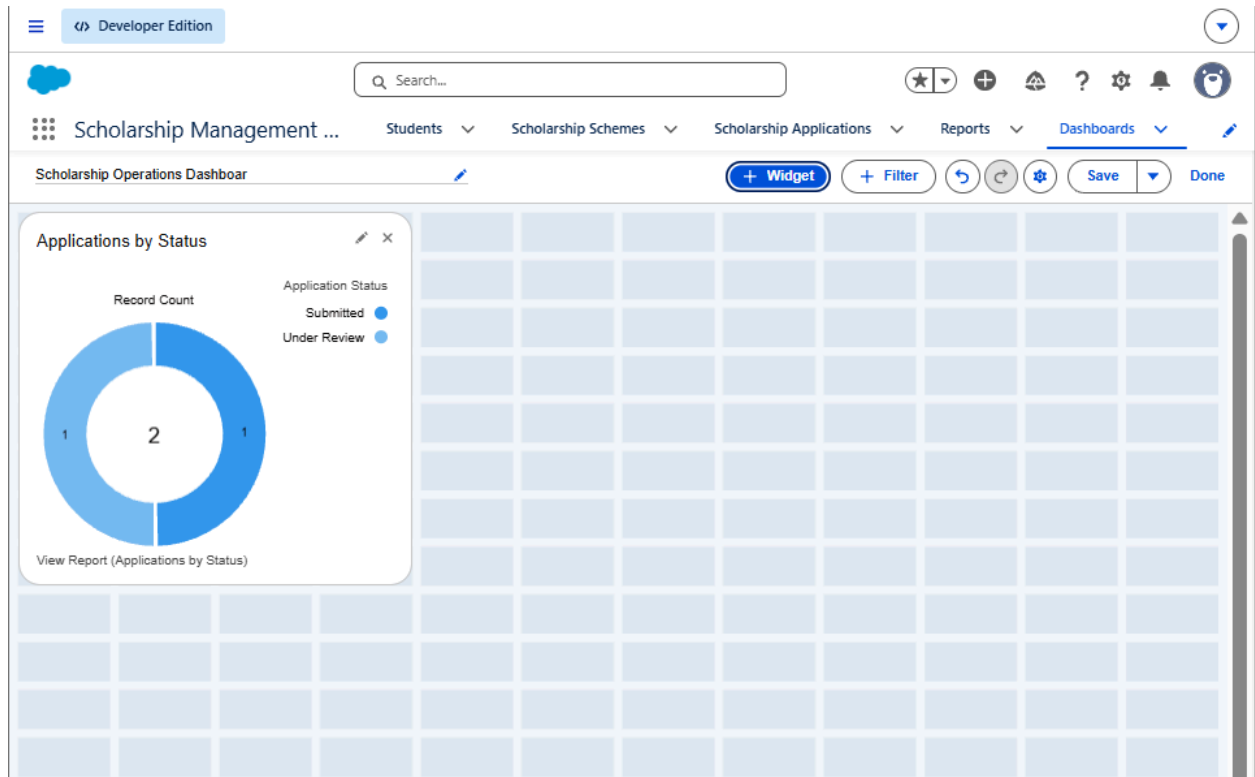


Explanation: A custom summary report titled Applications by Status was configured using the standard report type **Students with Scholarship Applications and Scholarship Scheme**.

- **Row Grouping:** Grouped by **Application Status** to measure pipeline volumes.
- **Filters:** Set to **All scholarship applications across All Time**.
- **Visualization:** Generates an integrated column visualization for immediate insight into submissions.

8. Executive Monitoring Dashboard





Explanation: The Scholarship Operations Dashboard provides visual operational oversight for institutional reviewers.

- **Widget Structure:** Real-time Donut Chart displaying the distribution of applications across statuses (Submitted, Under Review, Approved).
- **Operational Value:** Facilitates faster review turnaround times, identifies processing bottlenecks, and displays application distributions without manual tallying.

Done by

DS MADHUMITHA

23ISR028